Maths Exploring for Exercises at Terminal-06

Progressive exercises

E1 · Direct application. We define on $\mathbb{R}$ functions $f(x) = (2x-1)^5$ and $g(x) = e^{-x^2}$. Calculate $f'(x)$ and $g'(x)$.

E2 · Application and justification. Let f the function defined on $\mathbb{R}$ by $f(x) = x^4 - 6x^2$. Determine the intervals over which f is convex or concave.

E3 · Method to be organised. Note $\mathcal{C}$ the exponential function curve. Determine an equation
1. With the help of convexity, deduct from it an inequality valid for
all real x.

E4 · Choice of strategy. Demonstrate that the equation $x^3 + 2x = 5$ admits a single real solution.
Frame this solution between two consecutive integers.

E5 · Synthesis or demonstration. A dichotomy algorithm starts from the interval $[0;2]$ and divides its length by two at each step. Determine the minimum number of steps ensuring an interval of a length not exceeding 10^{-3}.